RACHEL POWER BI ASSIGNMENT-2

Creating Professional Dashboard for Electrical Vehicles Energy Efficiency

Summary:

Through this Dashboard we can analyse different types of Electrical vehicles, Vehicle Class and the different Models Manufactured by Individual Company such Audi, BMW, Mercedez-Benz, Rolls Royce etc. and through this interactive dashboard We can Analyze How Efficient the EV could travel based on their Recharge Time(h), Motor(KW) and its Energy Efficiency(km/kWh) throughout different Manufactured years. We can also observe and compare the old Model and New Model EV's performance based on Manufacturing year.

Dataset:

I took EV Energy Efficiency Dataset preprocessed dataset for creating Professional Dashboard. It consists of Following Features:

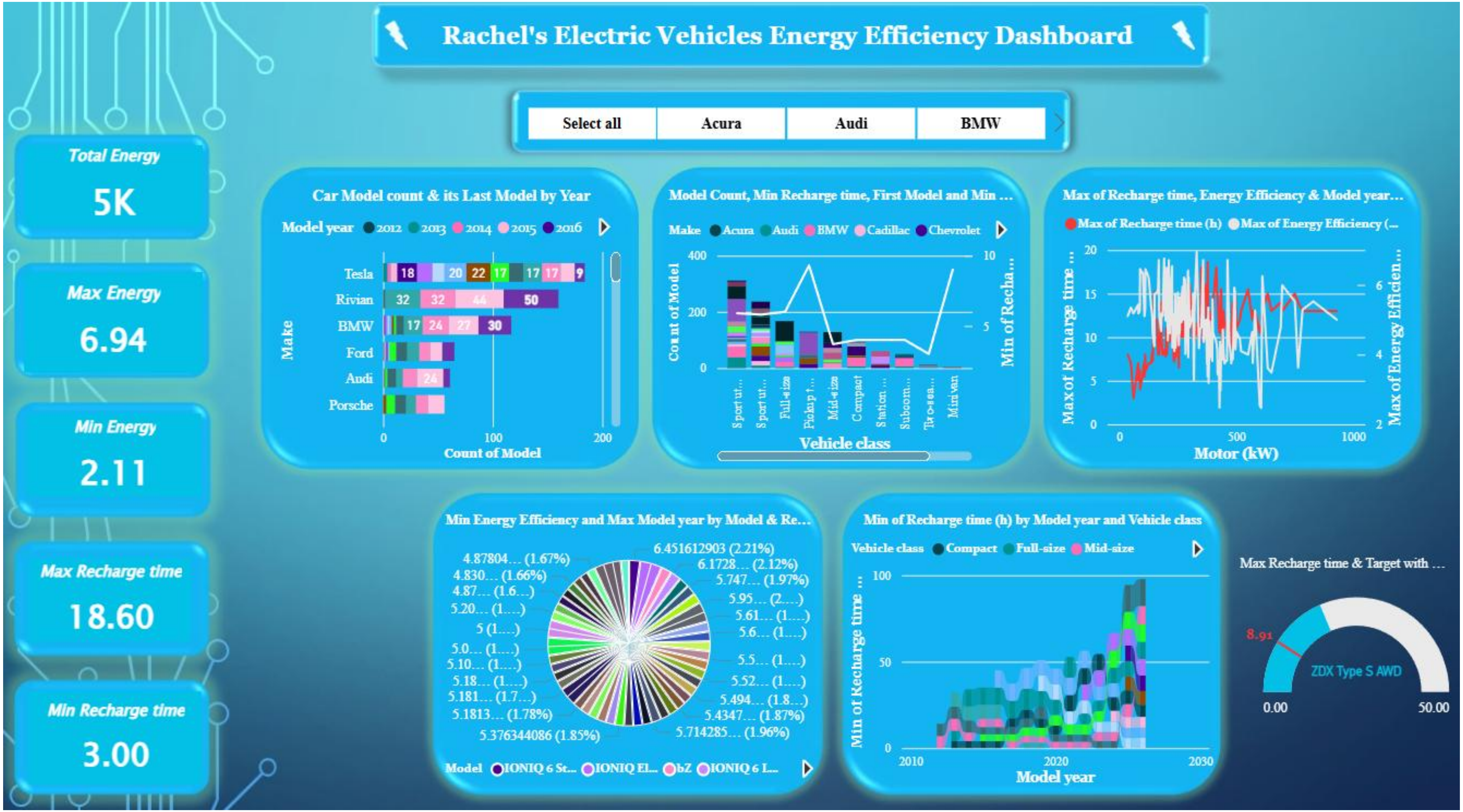
- Model year – Manufacturing year
- Make - Vehicle Manufacturer
- Model - Specific Vehicle Model
- Vehicle Class – Size Category
- Motor(kW) – Motor Power Rating
- Recharge Time(h) – Full Recharge Duration
- Energy Efficiency (km/kWh) – Target: Distance per kWh

Tool:

- Power BI

Following Analysis Done:

- **Card Charts –** Total Energy Efficiency, Max and Min Energy Efficiency, Min and Max Recharge Time.
- **Stack Bar Chart –** Vehicle Manufacturer, Count of Vehicle Models and its Last model by year
- **Line and Stacked Column Chart -**Vehicle Manufacturer and Specific type Model Count, Min Recharge time, First Model and Min of Model year
- **Line Chart -** Max of Recharge time, Max Energy Efficiency & Max of Model year by Motor (kW)
- **Pie Chart -** Min Energy Efficiency and Max Model year by Model & Recharge time (h)
- **Ribbon Chart -** Min of Recharge time (h) by Model year and Vehicle class
- **Gauge Chart -** Maximum Recharge time and Target with Last Model
  Target = AVERAGE('EV Energy Efficiency Dataset'[Energy Efficiency (km/kWh)])*2


Rachel's Electric Vehicles Energy Efficiency Dashboard
Select all
Acura
Audi
BMW
Total Energy
5K
Max Energy
6.94
Min Energy
2.11
Max Recharge time
18.60
Min Recharge time
3.00
Car Model count & its Last Model by Year
Model year 2012 2013 2014 2015 2016
Make
Tesla
Rivian
BMW
Ford
Audi
Porsche
Count of Model
Model Count, Min Recharge time, First Model and Min ...
Make Acura Audi BMW Cadillac Chevrolet
Count of Model
Min of Recha...
Vehicle class
Max of Recharge time, Energy Efficiency & Model year...
Max of Recharge time (h) Max of Energy Efficiency (...
Max of Recharge time ...
Max of Energy Efficien...
Motor (kW)
Min Energy Efficiency and Max Model year by Model & Re...
6.451612903 (2.21%)
5.376344086 (1.85%)
Model IONIQ 6 St... IONIQ El... bZ IONIQ 6 L...
Min of Recharge time (h) by Model year and Vehicle class
Vehicle class Compact Full-size Mid-size
Min of Recharge time ...
Model year
Max Recharge time & Target with ...
8.91
ZDX Type S AWD
0.00
50.00